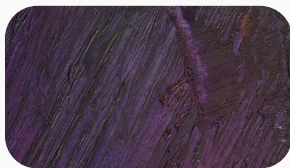
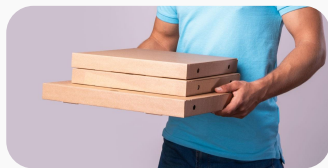
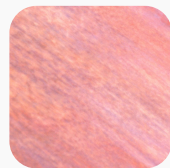
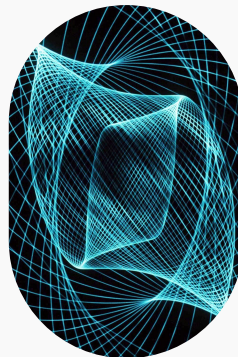


Quantum-Assisted Portfolio Co-Pilot for Guardrail-Constrained Multi-Asset Allocation



The Problem

Allocate capital across equities, fixed income, commodities, currencies, and alternatives to maximise expected utility, while honoring tunable goals and hard guardrails, and staying interpretable enough to explain to an investment committee.



Inputs

Return, risk, cost forecasts
+ tunable goals



Optimisation Solver

Constraints + objective
Classical + Quantum

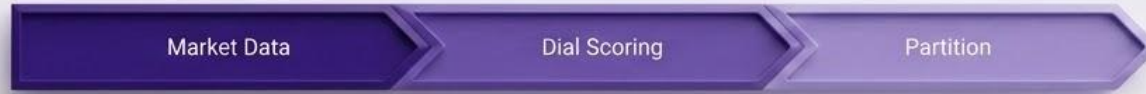


Outputs

Allocation +
explainable rationale

Where quantum actually helps: Not the whole portfolio, and not by replacing classical optimisation, but by handling the genuinely combinatorial middle where quantum classical convex methods have to approximate.

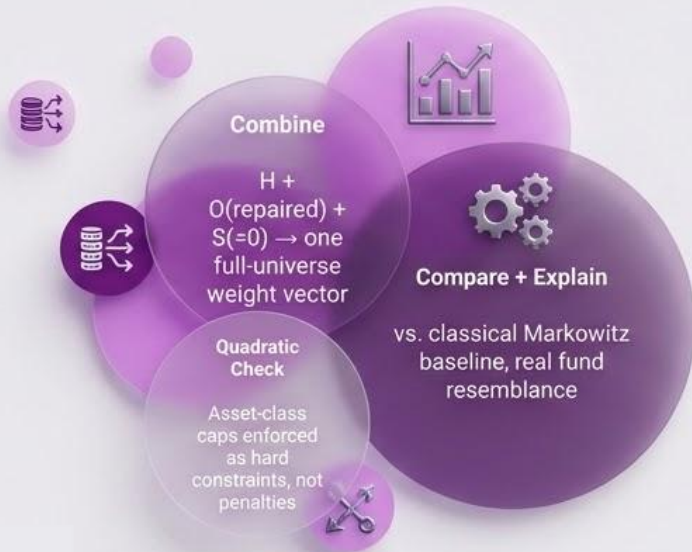
Architecture: The H/O/S Pipeline



- Real (yfinance+HF/DuckDB) or synthetic, real FRED rf, real Vanguard calibration

- Growth / Income / Drawdown / Cost – z-scored, combined per asset

- H / O / S split by conviction score



H – Hold
Classical convex QP · locked weights · $\leq 70\%$ budget



O – Optimise
QUBO → warm-start multi-restart QAQA · quantum-only step



S – Skip
Excluded entirely – weight forced to zero

Formulation

Decision variables

Each O-set asset's weight is binarised into b bits:
 $w_i = (\text{cap} / (2^b - 1)) \cdot \sum_k 2^k \cdot b_{\{i,k\}}$

2^b discrete levels per asset, capped by construction
— not a penalty term.

Objective (maximise)

$$\mu^T w - \lambda \cdot w^T \Sigma w - \text{cost}^T w$$

return risk (full covariance) cost

Constraints

- Budget equality: $\Sigma w = \text{O-budget}$ (leftover after H locked)
- Class-cap inequalities — added only where a class could bind, to avoid burning qubits on slack that can't matter

Why this is quantum-compatible

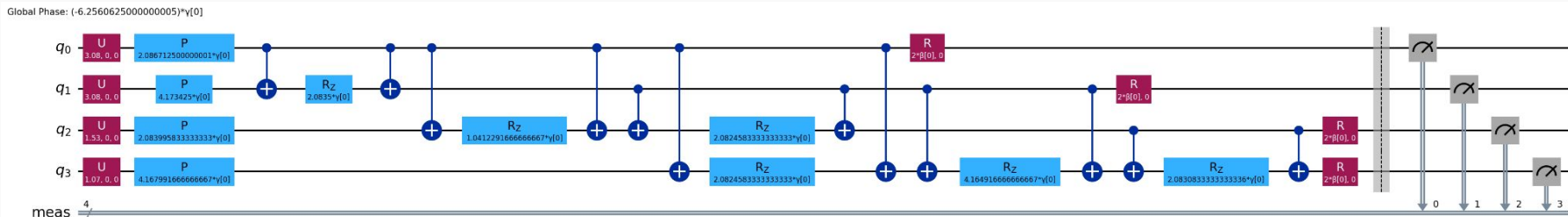
Built via `qiskit-optimisation`'s `QuadraticProgram`, converted with `QuadraticProgramToQubo` — the standard, auditable route from a constrained problem to an unconstrained QUBO. Equality constraints become penalty terms; inequalities get slack variables automatically, rather than hand-rolled penalty engineering.

Only the O-set — the genuinely contested middle of the portfolio — ever reaches this formulation. H is solved as a real convex QP (provably optimal for its own membership); S is never even considered.

Total qubit count = objective vars + slack vars. In this project, that varied 12–21 qubits for the same O-set size — slack depends on which class caps are binding, not on O-set size alone.

Quantum Solver

Warm-Start, Multi-Restart QAOA



Illustrative small instance (2 assets, 4 qubits, no slack). Production solves run 12–21 qubits.

1. Warm-start bias

Ry rotations from a classical relaxation of the QUBO — biases the initial state toward a good region instead of uniform superposition.

2. Cost Hamiltonian

CNOT–RZ–CNOT ladders encode the portfolio's real covariance matrix as two-qubit ZZ interactions — γ layer.

3. Mixer + measure

Standard X-mixer (β layer), then measurement. Multi-restart (best-of-N) guards against a single bad optimization run.

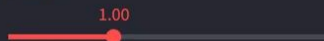
Why not XY-mixer / Dicke-state constrained mixers (the Week-2 literature-review candidate)? That family fits cardinality-selection (choose exactly K of N) — this is binarised continuous weights, a different encoding.

Tunable Goals → Instant Partition

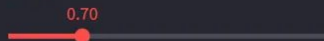
Tunable Goals

The four dials named in the brief.
Growth/income/drawdown/cost.

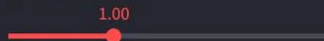
Growth



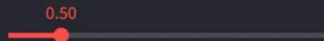
Income



Drawdown Control



Cost Sensitivity



> Advanced

Asset Scores & Partition

Updates instantly as dials move — this is the fast classical layer.



Locked (H) Allocation

5 assets, provably optimal for this membership (convex QP, not a heuristic).

	0
SPY	25.0%
LQD	25.0%
JNK	14.3%
IWM	5.7%

H budget used ⓘ

70.0%

Remaining O-set budget

30.0%

Four dials from the brief:
growth, income, drawdown control, cost sensitivity, drive a live conviction score, re-partitioned into H/O/S in under a second.

Tunable Goals → Instant Partition

Tunable Goals

The four dials named in the brief.
Growth/income/drawdown/cost.

Growth

1.00

Income

0.70

Drawdown Control

1.00

Cost Sensitivity

0.50

> Advanced

QAOA matched the exact optimum on this O-set instance.

Recommended Allocation vs. Classical Baseline

Expected Return

5.46%

↓ -1.79% vs Markowitz

Risk (Variance)

0.0145

↑ 0.0086 vs Markowitz

Resembles (real fund) ?

VSCGX

Markowitz baseline: Sharpe 0.438 (rf=3.90%, live FRED) — resembles LifeStrategy Conservative Growth (VSCGX)



Real market data run (Colab, **live yfinance + FRED**): QAOA matched the exact brute-force optimum on this O-set instance, and the full comparison — return, risk, real-fund resemblance — renders live.

Validation vs Classical

Equal-Footing Classical Benchmarks

ILP, greedy, and random all solving the **IDENTICAL** discretised O-set problem QAOA solves. Same bit-encoded weights, same budget/class-cap constraints. Not a looser comparison; the *exact same* problem instance.

Method	Objective	Gap to exact	Time (s)
Exact (brute force)	0.018763	0.000000	0.0000
QAOA (warm-start + multi-restart)	0.018763	0.000000	0.0000
ILP (diagonal risk)	0.018218	-0.000544	0.48
Greedy	0.018218	-0.000544	0.0005
Random (best-of-500)	0.018763	0.000000	0.08

QAOA's edge over ILP/greedy comes specifically from using the FULL quadratic risk term (real cross-asset covariance) — which the classical ILP had to drop (diagonal-only approximation) to stay a genuinely solvable linear program. A fair, non-cherry-picked reason for an advantage, not an artificially weakened baseline.

Scaling

Matrix Product States vs. Partitioning

27

Qubit Ceiling

AerSimulator's statevector target
(Circuit Too Wide For Target at 28),
independent of MPS

0.31s $\rightarrow$ 7.87s

Uncapped Runtime

p=1 $\rightarrow$ 2 $\rightarrow$ 3 at fixed 15 qubits, scales
sharply with circuit depth, not qubit
count

21 qubits

Exact Match

uncapped MPS matched statevector
exactly at every size tested, correctness
held throughout

What actually drives MPS's benefit?

- **Entanglement structure determines fit:** A purely random dense Hamiltonian choked MPS near 20 qubits, but covariance-structured QUBO (real assets) scaled cleanly through 21.
- **Opposing forces:** MPS's qubit-count advantage and QAOA's depth requirement work against each other. Deeper circuits generate more entanglement, eroding MPS's efficiency edge.
- **Implementation fix:** A cross-backend bitstring-labelling bug was resolved along the way by comparing Hamiltonian expectation values instead of decoded bitstrings.

Reliability

Multi-Seed Variance: Reliability Isn't Free at Scale

12 qubits

12/12

seeds matched the exact brute-force optimum

Zero variance. The warm-start + multi-restart fix (replacing the earlier single-seed random-init, which failed catastrophically 1/7 times) appears to fully work at this size.

18 qubits

0/10

seeds matched the exact optimum but worst-case gap only 9.8%

Real, qualitative shift. Not catastrophic (no guardrail breach, no wild outlier), but exactness is no longer guaranteed. Run at reduced settings (n_restarts=1 vs. production's 3), likely an overestimate of production variance.

Multi-restart QAOA's reliability is not a fixed property of the method. It degrades as problem size grows. This directly updates the open question from earlier development: the fix works, but not uniformly.

Reliability

Walk-Forward: 14 Periods, 2010–2026

H/O/S + QAOA

44.8%

cumulative return • mean 3.11%/period • std 9.35%

Markowitz baseline

21.5%

cumulative return • mean 1.97%/period • std 10.51%

Zero guardrail breaches: either method, every single one of 14 periods, across 16 years of rolling re-optimisation.

Important Note

9/14 periods beat Markowitz (64.3%) and QAOA matched the exact optimum in 10/14, but this ran single-seed, reduced QAOA settings, on 14. The outperformance is real for this run, not evidence QAOA reliably beats Markowitz in general. The zero-breach result is the more defensible claim, given how the rubric actually scores.

Why This Satisfies Constraints

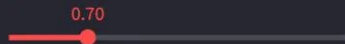
Tunable Goals

The four dials named in the brief.
Growth/income/drawdown/cost.

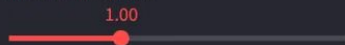
Growth



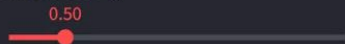
Income



Drawdown Control



Cost Sensitivity



> Advanced

Why This Satisfies Constraints

Asset Class	Used	Cap	Headroom
Equities	41.4%	80.0%	38.6%
Fixed Income	39.3%	80.0%	40.7%
Commodities	17.9%	20.0%	2.1%
Currencies	0.0%	15.0%	15.0%
Alternatives	0.0%	20.0%	20.0%

Zero guardrail breaches — every asset-class cap is a hard constraint the solver enforces.

H / O / S Rationale

- **H — Hold (5 assets):** top-scoring under the current dial settings, locked at a classically-optimized weight (a real convex QP solved for this exact membership, not a proportional-to-score heuristic) up to 70% of the portfolio.
- **O — Optimize (4 assets):** the contested middle — assets whose ranking is genuinely ambiguous given the current dials. This is the only part of the portfolio the quantum solver touches.
- **S — Skip (6 assets):** lowest-scoring under the current dial settings, excluded entirely (weight = 0).

Every asset-class cap is a hard constraint the solver enforces, not a penalty to trade off. Headroom shown explicitly, not just pass/fail.

Looking Ahead



Validation rigour

- **Benchmark Expansion**

Expand suite across more O-set instances with random seeds.

- **Backtest Constraints**

Add transaction-cost returns and explicit constraints to backtest.

- **Classical Comparators**

Add MIQP and nonlinear solvers as comparators.



Quantum scalability

- **Loop Integration**

Integrate MPS into QAOA loop to bypass statevector limits.

- **Depth Comparison**

Re-run depth scaling budget with parameter count (2p).

- **Variance Study**

Push multi-seed study with production QAOA settings.



Toward a real implementation

- **Advisor Workflow**

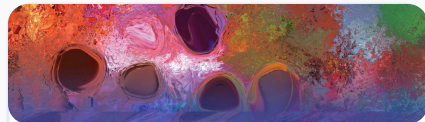
Evolve co-pilot to use approved fund universes and audited caches.

- **Human-in-the-Loop**

Enforce human approval before any portfolio trade.

- **Guardrails & Logs**

Add rich sector limits, compliance review, and reproducible run logs.



Submitting to arXiv

- ✓ Partitioning ablation
- ✓ Scale to larger sizes
- ✓ Statistical significance
- ✓ Formalise feasibility
- ✓ Turnover reporting
- ✓ Production confirmation

THANK YOU!

References:

Quantum & Classical methods

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2. Qiskit Optimization documentation, Quadratic Program To Qubo -- the constrained-to-QUBO conversion route used throughout (quantum/[qubo.py](#)).
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2. Vanguard LifeStrategy fund family fact sheets (fund numbers 0122, 0724, 0723, 0914), fetched directly from workplace.vanguard.com, June 30 2026 -- real fund-family stock/bond splits used to calibrate ASSET_CLASS_CAPS and the "resembles Vanguard's own fund" explainability feature (market_data/vanguard_calibration.py).
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4. defeatbeta/yahoo-finance-data dataset on Hugging Face -- real historical OHLCV for 9 of 15 tickers, queried via DuckDB predicate pushdown (market_data/loader.py).